llmXive Automated Review of SkillOpt: Executive Strategy for Self-Evolving Agent Skills

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The manuscript makes several strong factual claims regarding novelty and performance that are not fully supported by the visible text or citations.

First, the Introduction (Section 1) asserts that SkillOpt is "to our knowledge, the first systematic controllable text-space optimizer for agent skills." This claim is contradicted or at least heavily qualified by the Related Work section, which cites GEPA (Agrawal et al., 2025) and TextGrad (Yuksekgonul et al., 2024). GEPA is explicitly described as using "trajectory feedback to guide reflective prompt evolution," and TextGrad is a known text-space optimizer. The authors must clarify the specific definition of "controllable" that distinguishes their work from these prior art, or soften the "first" claim to avoid overstatement.

Second, the claim in Section 4.1 that the method is "best or tied on all 52 evaluated cells" is not fully substantiated in the provided text. Table 1 (Main Results) in the snippet only displays data for GPT-5.5 and GPT-5.4. The text mentions "seven target models," but the data for the remaining five models is omitted from the visible LaTeX source. Without seeing the full table or a summary of the other models, the specific count of "52 cells" cannot be verified.

Third, the transfer results in Section 4.3 cite specific gains (e.g., "+5.6 points on LiveMath" for GPT-5.4-nano) and claim these surpass in-domain references. However, the provided snippet of Table 2(a) (Cross-model transfer) only contains rows for SpreadsheetBench. The specific data points for LiveMath transfer are missing from the visible text, rendering the claim unverifiable in the current context.

Finally, the bibliography relies heavily on papers dated 2026 and model releases (GPT-5.5, GPT-5.4) that appear to be future-dated relative to the current real-world timeline. While this is a preprint, the validity of the "state-of-the-art" comparisons depends entirely on the acceptance of these future-dated references as established baselines. The paper should explicitly address the temporal context of these citations to ensure the claims of superiority are not based on hypothetical or unreleased benchmarks.

Action items.

- [**writing**] Section 1 claims SkillOpt is the 'first systematic controllable text-space optimizer.' This absolute claim is unsupported given the citation of GEPA (Agrawal et al., 2025) and TextGrad (Yuksekgonul et al., 2024), which also perform text-space optimization. The authors must qualify this claim (e.g., 'first to combine...') or provide evidence that prior works lack 'controllability' in the specific sense defined.

- [**writing**] Table 1 and Section 4.1 state SkillOpt is best or tied on 'all 52 evaluated cells.' However, Table 1 (e000) only displays results for GPT-5.5 and GPT-5.4. The text mentions 'seven target models' but does not list the other five or show their data in the provided snippet. The claim of 52 cells cannot be verified from the visible evidence; the full table or a summary of all 7 models is required to support this specific number.

- [**writing**] Section 4.3 claims a GPT-5.4 skill transfers to GPT-5.4-nano with a +5.6 point gain on LiveMath, 'surpassing the in-domain reference.' The text cites Table 2(a) for this, but the provided snippet of Table 2(a) only shows SpreadsheetBench results. The specific LiveMath transfer data point (+5.6) is missing from the visible text, making the claim unverifiable in the current source.

- [**writing**] The bibliography lists multiple 2026 papers (e.g., SkillsBench, SoK, EvoSkill) and future-dated model releases (GPT-5.4, GPT-5.5, Qwen3.6). While this is a preprint, the reliance on 'future' citations for baseline comparisons and the definition of 'state-of-the-art' requires a clear statement in the limitations or experimental setup regarding the temporal validity of these references.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively visualizes the optimization landscape and the selection gate concept, but the caption contains a missing system name and the diagram omits the specific optimizer model component mentioned in the text.

Figure 2

The figure provides a clear and detailed visual pipeline of the system architecture, but the caption contains a grammatical error where the subject name is missing.

Figure 3

Figure 3 presents performance trends but suffers from ambiguous axis scaling and unclear definition of the 'Selection best' metric, which undermines the interpretation of how validation performance guides checkpoint selection.

Action items.

- [**writing**] Figure 1: The caption contains a grammatical error ('Overview of .') where the system name (likely 'SkillOpt') is missing.
- [**science**] Figure 1: The diagram illustrates the 'Text-space optimization analogy' and the 'held-out selection gate' mechanism conceptually, but it does not depict the 'frontier optimizer model' or the 'trajectory' processing steps explicitly described in the caption.
- [**writing**] Figure 2: The caption reads 'Pipeline of .' with a missing noun (likely 'SkillOpt') immediately following the preposition.
- [**science**] Figure 3: The legend defines 'Selection best' as a single line, but the caption

describes it as a 'selection-best score on the validation set' used to pick a checkpoint. The plot shows this metric evolving over epochs, implying it is the score of the model at that epoch on the validation set, not a 'best' score. This creates ambiguity: is the orange line the validation score of the current epoch, or the running maximum? If it is the running maximum, it should be non-decreasing, but in (b) it app

- [**writing**] Figure 3: The x-axis label 'Epoch checkpoint' is ambiguous. In (a), checkpoints are 1,2,4,8; in (b) and (c), they are 1,2,4,8,12,16. The spacing on the axis is not uniform (e.g., distance from 1 to 2 equals 2 to 4, which equals 4 to 8), suggesting a log scale, but the tick labels are not formatted as such (e.g., 10^0, 10^1) and the axis is not labeled as logarithmic. This misrepresents the temporal progression of training.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript exhibits significant jargon density that may hinder accessibility for non-specialist readers, particularly those unfamiliar with reinforcement learning (RL) and deep learning optimization terminology.

In Section 3.1, the term 'harness' is introduced as variable 'h' without definition. While context suggests it refers to the execution environment, this term is not standard outside specific agent frameworks and should be explicitly defined as 'execution harness' or 'task execution environment' at first use.

Section 3.2 relies heavily on 'rollout', a term deeply rooted in RL that may confuse readers from other ML subfields. The phrase 'rollout batch' appears multiple times without explanation. Replacing this with 'execution batch' or 'task attempt batch' would improve clarity while maintaining precision.

Section 3.3 introduces 'minibatch reflection', combining standard ML terminology ('minibatch') with a novel concept ('reflection') without sufficient explanation. The paper should clarify that this refers to analyzing groups of execution traces to identify failure/success patterns, rather than assuming readers understand the 'reflection' metaphor.

Section 3.4's 'textual learning-rate' is a creative analogy but risks confusion. While the paper explains the analogy to learning rate, the term itself is jargon-heavy. The primary term should be 'edit budget' with 'textual learning-rate' as a parenthetical analogy for those familiar with optimization.

Section 3.5's 'slow/meta update' terminology is particularly opaque. The distinction between 'slow' (epoch-level) and 'meta' (optimizer-side) updates is not immediately clear to non-specialists. The paper should explicitly state that 'slow update' refers to longitudinal guidance written at epoch boundaries, while 'meta update' refers to optimizer-side memory that guides future edit generation.

Additionally, ‘trajectory’ appears frequently (Equation 1, Section 3.2) as an RL term for execution sequences. While standard in RL literature, replacing this with ‘execution sequence’ or ‘task history’ would make the paper more accessible to a broader audience without losing precision.

The ‘validation gate’ in Section 3.5 is another jargon term that could be simplified to ‘validation filter’ or ‘acceptance check’ for better accessibility.

These changes would significantly improve the paper’s accessibility while maintaining its technical precision, aligning with the goal of making advanced research accessible to a broader scientific community.

Action items.

- [**writing**] Define ‘harness’ at first use in Section 3.1. Currently, ‘h’ is introduced as a variable without explaining that it represents the execution environment or tool interface, which may confuse non-specialist readers.”
- [**writing**] Replace ‘rollout’ with ‘execution trace’ or ‘task attempt’ in Section 3.2 and throughout. ‘Rollout’ is RL jargon that obscures meaning for readers unfamiliar with reinforcement learning terminology.”
- [**writing**] Define ‘minibatch reflection’ in Section 3.3. The term combines ML ‘minibatch’ with a novel concept ‘reflection’ without clarifying that this refers to analyzing groups of execution traces to identify patterns.”
- [**writing**] Clarify ‘textual learning-rate’ in Section 3.4. While the analogy to learning rate is explained, the term itself is jargon-heavy. Consider ‘edit budget’ as the primary term with ‘textual learning-rate’ as a parenthetical analogy.”
- [**writing**] Define ‘slow/meta update’ in Section 3.5. The distinction between ‘slow’ and ‘meta’ updates is unclear to non-specialists. Explain that ‘slow’ refers to epoch-level guidance and ‘meta’ refers to optimizer-side memory.”
- [**writing**] Replace ‘trajectory’ with ‘execution sequence’ or ‘task history’ in Equation 1 and Section 3.2. ‘Trajectory’ is standard RL jargon that may not be immediately clear to all readers.”
- [**writing**] Define ‘validation gate’ in Section 3.5. While the concept is described, the term ‘gate’ is jargon. Consider ‘validation filter’ or ‘acceptance check’ for broader accessibility.”

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The manuscript presents a coherent logical framework for SkillOpt, where the central premise—that a frozen agent’s performance can be improved by optimizing a separate, bounded text-space skill document via a validation gate—is well-supported by the described methodology. The causal chain from trajectory analysis to edit proposal, validation gating, and slow updates is internally consistent.

However, there are specific logical gaps regarding the aggregation of experimental results and the attribution of ablation effects. First, the repeated claim of "52 of 52 evaluated cells" (Introduction, Section 5.1, Conclusion) lacks a clear derivation from the stated experimental setup. The paper mentions "six benchmarks, seven target models, and three execution harnesses," which would imply 126 cells (6×7×3), yet the claim is 52. Without a clear definition of which specific combinations constitute these 52 cells, the absolute claim of "best or tied on all" is logically unsupported by the provided text.

Second, the ablation results in Table 3 and Section 5.2 present a potential causal ambiguity. The authors state that the "meta skill" is "optimizer-side only and is not shipped with the target model" (Section 3.6), yet the ablation removing "both meta skill and slow update" results in a massive performance drop (22.5 points). If the meta skill is not part of the final artifact, its removal should not directly impact the test-time performance of the *final* skill unless it critically influences the *generation* of the slow update or the edit proposals during training. The text conflates the removal of the "meta skill" (a training-time guide) with the removal of the "slow update" (a structural component of the skill), making it difficult to logically isolate which component is responsible for the observed performance degradation.

Finally, the cross-harness transfer results (Section 5.3) claim that a skill trained in the Codex harness outperforms a skill trained natively in the Claude Code harness. While the "harness-agnostic" design is a strong premise, the logical mechanism for why a skill optimized for one specific tool-use environment (Codex) would generalize *better* than one optimized for the target environment (Claude Code) is not fully elaborated. The paper asserts this happens but does not sufficiently explain the logical bridge between the "bounded text-space" abstraction and the specific execution nuances of different harnesses that would allow for such superior transfer.

Action items.

- [**writing**] The manuscript presents a coherent logical framework for SkillOpt, where the central premise—that a frozen agent's performance can be improved by optimizing a separate, bounded text-space skill document via a validation gate—is well-supported by the described methodology. The causal chain from trajectory analysis to edit proposal, validation gating, and slow updates is internally consistent. However, there are specific logical gaps regarding the aggregation of experimental results and the attrib

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The manuscript makes several claims that extend beyond the immediate evidence provided, particularly regarding novelty and the universality of performance gains.

First, the Introduction asserts that SkillOpt is, "to our knowledge, the first systematic controllable text-space optimizer for agent skills." This is an overreach. The Related Work section and the bibliography explicitly cite GEPA (trajectory feedback for prompt evolution), TextGrad (automatic differentiation via text), and EvoTest (evolutionary test-time learning). While SkillOpt introduces specific mechanisms like bounded edits and a strict validation gate, claiming to be the "first" in the broad category of text-space optimization ignores the direct lineage of these works. The authors should refine this claim to highlight the specific *constraints* (e.g., bounded add/delete/replace) or the *validation gating* as the novel contribution, rather than the general concept of text-space optimization.

Second, the paper repeatedly emphasizes that the method is "best or tied on all 52 evaluated cells" (Introduction, Section 5.1). This is a definitive statistical claim. However, Table 1 and the text do not report standard deviations, confidence intervals, or p-values for these 52 measurements. In the absence of variance data, the claim of being "tied" is ambiguous (is it within noise margin or exactly equal?), and the claim of being "best" lacks statistical significance testing. The authors must provide error bars or statistical tests to support the assertion that these results are robust and not due to random fluctuation, especially given the high variance often seen in LLM benchmarks.

Third, the cross-harness transfer results (Table 3b) show a +59.7 point gain when transferring a skill from Codex to Claude Code. While impressive, the paper does not sufficiently analyze whether this transfer is due to genuine procedural generalization or if the skill is simply exploiting a shared formatting quirk or evaluation artifact between the two harnesses. The "Limitations" section mentions that skills can encode domain-specific heuristics, but the main text treats the transfer as a robust proof of generalizability without ruling out harness-specific overfitting. A deeper analysis of *why* the transfer works (e.g., comparing the actual skill text applied in both contexts) is needed to justify the magnitude of the claim.

Finally, the claim that the method works "across six benchmarks, seven target models, and three execution harnesses" (Introduction) is supported by the tables, but the "seven target models" claim is slightly obscured. The tables list GPT-5.5, GPT-5.4, and their mini/nano variants, but the text should explicitly enumerate all seven to ensure the reader can verify the scope of the claim.

Action items.

- [**writing**] The claim of being the 'first systematic controllable text-space optimizer' (Introduction) is overreaching given the existence of GEPA, TextGrad, and EvoTest which also perform trajectory-guided text optimization. The authors must clarify the specific novelty of their 'bounded edit' and 'validation gate' mechanisms relative to these baselines rather than asserting a broad 'first' status.
- [**science**] The statement that SkillOpt is 'best or tied on all 52 evaluated cells' (Introduction, Section 5.1) is a strong statistical claim that requires explicit reporting of variance (standard deviation or confidence intervals) for every cell in Table 1. Without this, the 'tied' claim is ambiguous and the 'best' claim lacks statistical rigor.

- [**science**] The cross-harness transfer claim of '+59.7 points' (Table 3b, Section 5.3) implies a massive generalization leap. The paper must explicitly discuss whether the 'Codex' and 'Claude Code' harnesses share underlying evaluation logic or if the skill is merely exploiting a specific quirk of the target harness's prompt format, rather than true procedural transfer.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a method for optimizing agent skills via text-space editing. From a safety and ethics perspective, the current draft is largely sound in its technical description but lacks necessary disclosures regarding potential misuse and data governance.

First, the **dual-use potential** is not adequately addressed. The method optimizes skills for tasks like spreadsheet manipulation (SpreadsheetBench) and embodied decision-making (ALFWorld). While the paper claims the skills are "procedural," there is no discussion on whether the optimization process could inadvertently discover and encode strategies for bypassing safety protocols, exploiting software vulnerabilities, or automating malicious tasks. The "Limitations" section (e001) mentions reliance on automatic verifiers but fails to explicitly state that the system is not designed to optimize for adversarial or harmful outcomes. A dedicated paragraph in the Limitations or a new "Safety Considerations" section is required to discuss these risks and the authors' stance on preventing misuse.

Second, **data privacy and consent** are not explicitly covered. The optimization loop (Algorithm 1, e001) utilizes training splits from various benchmarks (e.g., SearchQA, OfficeQA). The paper does not state whether these datasets contain personally identifiable information (PII) or copyrighted content. Given that the "self-evolving" agent learns from trajectories, there is a risk that the final skill artifact could memorize and reproduce sensitive data from the training distribution. The authors must include a statement confirming that the datasets used are public, consented, and free of PII, or describe any preprocessing steps taken to sanitize the data.

Finally, the **validation gate** (Section 2.5, e000) is described as strictly accepting edits that improve a score. However, the definition of "score" is critical. If the evaluation harness rewards efficiency or task completion without explicit safety constraints, the optimizer might prioritize harmful shortcuts. The paper should clarify if safety constraints are integrated into the reward function or if the validation gate includes a safety check to reject skills that exhibit adversarial behavior, even if they improve the primary task metric.

These additions are necessary to ensure the research is responsibly communicated and to mitigate potential downstream harms.

Action items.

- [**writing**] The 'Limitations' section (e001) acknowledges reliance on 'automatic verifiers' but lacks a specific discussion on dual-use risks. If the optimized skills encode heuristics for bypassing safety filters or exploiting software vulnerabilities (e.g., in SpreadsheetBench or ALFWorld), the paper must explicitly address these potential misuse vectors and propose mitigation strategies.

- [**writing**] The optimization loop (Algorithm 1, e001) relies on a 'held-out selection split' for validation. The manuscript does not specify the data provenance or consent status of the benchmarks used (e.g., SearchQA, OfficeQA). If any training data contains PII or copyrighted material, the 'self-evolving' nature of the skill could inadvertently memorize and reproduce sensitive information. A statement on data privacy and copyright compliance is required.
- [**writing**] The 'Optimizer Prompt Contracts' (e001) instruct the optimizer to propose edits based on 'failure patterns.' There is no explicit guardrail mentioned to prevent the optimizer from generating instructions that optimize for 'jailbreaking' or adversarial behavior if the evaluation harness inadvertently rewards such outputs. The paper should clarify if safety constraints are hard-coded into the reward function or validation gate.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The manuscript presents a compelling framework for optimizing agent skills via text-space editing, supported by extensive empirical results across six benchmarks. However, the scientific evidence supporting the magnitude and robustness of the claims requires strengthening in three key areas.

First, the paper lacks statistical rigor in reporting results. Table 1 and the main text report point estimates (e.g., "87.3" vs "77.7") but omit measures of variance (standard deviation) or statistical significance (p-values). Given that some reported gains are marginal (e.g., +0.2 to +1.3 points in cross-benchmark transfer), it is unclear if these improvements are statistically distinguishable from random noise or baseline variance. The claim of being "best or tied on all 52 cells" is strong; without significance testing, this could be an artifact of overfitting to the specific test split or random fluctuation.

Second, the sample sizes for the evaluation splits are not explicitly stated. While the 2:1:7 train/selection/test split ratio is mentioned in the appendix, the absolute number of test examples (N) for each benchmark (e.g., SearchQA, SpreadsheetBench) is missing. Without N, readers cannot calculate the standard error of the mean or assess the confidence intervals of the reported accuracy percentages. This is critical for evaluating the reliability of the "best or tied" claim.

Third, the ablation studies, while present, do not fully isolate the contribution of the core "minibatch reflection" mechanism from the general effect of search. The paper compares against "no-skill" and "human-skill" baselines but lacks a comparison against a simpler search strategy (e.g., random edits or greedy single-step edits) that does not use the complex reflection/merge pipeline. This makes it difficult to determine if the performance gains are due to the specific algorithmic novelty or simply the result of exploring a larger search space. The component ablation (Table 2) shows the value of the slow/meta update, but the baseline for the reflection mechanism itself remains somewhat opaque.

To address these concerns, the authors should report standard deviations over multiple runs (if applicable) or provide confidence intervals for the main results, explicitly state the test set sizes, and include a baseline that controls for search budget without the reflection logic.

Action items.

- [**science**] Report statistical significance (p-values, confidence intervals, or standard deviations) for the reported gains. The claim of being 'best or tied on all 52 cells' lacks evidence of whether differences are statistically significant or within noise margins, especially for small gains (e.g., +0.2 to +1.3 points).
- [**science**] Clarify the sample size (N) for the held-out test splits used in Table 1. The text mentions a 2:1:7 split but does not state the absolute number of test examples per benchmark, making it impossible to assess the statistical power of the reported accuracy percentages.
- [**science**] Provide a baseline comparison against a simple random-edit or greedy-search baseline to rule out that the gains are merely due to the search budget rather than the specific 'minibatch reflection' and 'slow/meta update' mechanisms.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The manuscript presents a comprehensive empirical evaluation of SkillOpt across six benchmarks and seven models. However, the statistical rigor of the reported results requires clarification to support the strong claims made.

First, the central claim that the method is "best or tied on all 52 evaluated cells" (Introduction, Section 5.1) relies entirely on point estimates. The paper does not report standard errors, confidence intervals, or variance across multiple runs for any of the reported percentages (e.g., 87.3% on SearchQA). Given the inherent stochasticity of LLM generation and the relatively small number of benchmarks (n=6), the absence of uncertainty quantification makes it impossible to determine if the observed "ties" or marginal wins are statistically significant or within the noise floor of the evaluation harness. The authors should report 95% confidence intervals or standard deviations for the main results in Table 1.

Second, the statistical testing methodology is not described. The ablation studies (Section 5.2, Table 3) report specific point-difference drops (e.g., -22.5 points for removing the slow/meta update). Without a stated statistical test (e.g., paired t-test or Wilcoxon signed-rank test across benchmarks or seeds), it is unclear if these differences are robust. The "52/52" claim implies a deterministic superiority that is statistically unlikely in LLM benchmarks without rigorous significance testing.

Third, the experimental protocol (Appendix E) mentions a “default 2:1:7 split” but does not explicitly state whether the reported results are averaged over multiple random seeds or if a single fixed seed was used. For reproducibility and to rule out selection bias in the test set, the authors must clarify the randomization strategy and, ideally, report results averaged over at least 3-5 seeds with variance.

Finally, the cost analysis (Table 4) presents “Cost / pt” as a deterministic ratio. Since the “pt” (point gain) lacks an uncertainty interval, the derived cost metric also lacks statistical grounding. The authors should ensure that all comparative claims are backed by appropriate statistical tests or uncertainty bounds.

Action items.

- [**science**] Report uncertainty estimates (e.g., 95% CIs or SE) for the 52 reported test scores. The claim of ‘best or tied’ on every cell is statistically fragile without variance measures, especially given the small number of benchmarks (n=6) and potential run-to-run variance in LLM agents.
- [**science**] Clarify the statistical significance of the reported gains (e.g., +23.5 points). With only 6 benchmarks, a simple mean comparison is insufficient. Specify if paired tests (e.g., Wilcoxon signed-rank) were used across benchmarks or if the ‘52/52’ claim relies on point estimates alone.
- [**science**] Define the randomization protocol for the train/selection/test splits. The text mentions a ‘default 2:1:7 split’ but does not specify if seeds were fixed or if results are averaged over multiple random seeds, which is critical for reproducibility in stochastic LLM evaluations.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a clear and generally well-structured argument for the SkillOpt framework. The introduction effectively sets up the problem of hand-crafted skills versus trainable external states, and the method section logically breaks down the optimization loop into forward, backward, and control components. The prose is generally concise, and the use of bolding for key metrics (e.g., “+23.5 points”) aids readability.

However, the writing quality is significantly impacted by structural redundancy. The Appendix (Section e001/e002) duplicates large portions of the main text, including the entire “Limitations” section, “Experimental Protocol Details,” and the “Optimization Procedure” with Algorithm 1. This repetition is unnecessary and disrupts the narrative flow, making the document feel bloated. The authors should consolidate these sections, ensuring that the main text contains the necessary details for a standalone read while the appendix serves only for extended technical proofs or raw prompt data not essential to the core narrative.

Additionally, there are minor inconsistencies in referencing and terminology. In Section 3.2 (Ablations), the text cites “Table~\\ref{tab:ablation_sweeps},” which does not appear in the provided source code; the reader is left without the referenced data. Furthermore, the model naming convention fluctuates between “GPT–5.5” and “GPT–5.4” in ways that do not always align with the provided bibliography keys (e.g., `openai2026gpt54`), creating potential confusion regarding which model versions were actually tested. Finally, while the LaTeX source uses double hyphens for en-dashes in model names (e.g., `GPT--5.5`), this is a stylistic choice that should be applied consistently if it is the intended format, but currently, it appears alongside standard hyphens in other contexts, suggesting a lack of global style enforcement. Addressing these structural and consistency issues will significantly improve the manuscript’s professionalism and readability.

Action items.

- [**writing**] The manuscript contains significant redundancy between the main body and the appendix. Specifically, the ‘Limitations’, ‘Experimental Protocol Details’, and ‘Optimization Procedure’ sections (including Algorithm 1) are duplicated verbatim in both Section 6 and the Appendix. This inflates the paper length and disrupts the reading flow. Please remove the duplicate content from the appendix or the main text, retaining only one complete version.
- [**writing**] In Section 3 (Method), the text references ‘Table~\ref{tab:ablation_sweeps}’ in the Ablations subsection, but this table is not present in the provided LaTeX source (only ‘tab:component_ablation’ and ‘tab:transfer_all’ are defined). Verify the correct label or add the missing table to ensure the reader can follow the argument.
- [**writing**] The phrase ‘GPT–5.5’ appears in the Introduction and Conclusion, while ‘GPT–5.4’ is used elsewhere. Ensure consistent naming conventions for model versions throughout the manuscript, as the double-hyphen usage varies or the version numbers seem inconsistent with the cited references (e.g., openai2026gpt54).